# Sam Guessoum

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## RESEARCH PROFILE

Artificial intelligence researcher and Computer Science PhD candidate at Worcester Polytechnic Institute with a 4.0/4.0 M.S. in Artificial Intelligence. Research spans biomedical AI, ophthalmic surgical-video analysis, multimodal decision support, applied optimization, and multi-agent systems, with accepted work across Computing in Cardiology, IEEE venues, AMAI, HECKTOR, AFRICAI/MICCAI, and Springer LNCS-related research.

## EDUCATION

**Worcester Polytechnic Institute** - PhD candidate in Computer Science; M.S. Artificial Intelligence, GPA 4.0/4.0 | PhD 2026 - 2029; M.S. Aug 2024 - Aug 2026

Thesis: Adaptive Multi-Agent Heuristic Orchestration for 1D/2D Bin-Packing Problems, extended toward industrial 3D bin packing.

**University of Greenwich** - B.Sc. Applied Mathematics | Sep 2020 - Aug 2023

Machine Learning examination: 85%; capstone on English Premier League match prediction.

## RESEARCH AND PROFESSIONAL EXPERIENCE

**Research Intern, Harvard Ophthalmology AI Lab, Massachusetts Eye and Ear / Harvard Medical School**

Sep 2026 - May 2027 | Supervisor: Mohammad Eslami

- Developing a cataract-surgery video intelligence project that converts surgical video into structured timelines, interactive reports, and evidence-linked review outputs for ophthalmology research.
- Own the end-to-end research workflow: data organization, model evaluation, experiment tracking, reviewer-facing figures, manuscript drafting, and coordination with hospital-affiliated collaborators.
- Reported strong locked-test-set results for phase recognition, instrument detection, and clinically interpretable timestamped outputs.

**Research Assistant, Worcester Polytechnic Institute**

Jan 2025 - Present | Advisor: Chun-Kit Ngan; collaborator: Rolf Bardeli

- Lead research on adaptive multi-agent optimization for 1D, 2D, and 3D bin-packing problems under academic and industrial constraints.
- Designed reproducible benchmark studies across 3,000+ problem instances and 12 datasets, comparing solution quality, runtime, scalability, and generalization across packing geometries.
- Contributed to a peer-reviewed bin-packing paper and Springer LNCS-related extension, reporting 9% average bin-utilization improvement, 10x speedup over Branch and Bound, and <=2% optimality gap.

**Research Collaborator, RISE Lab, Johns Hopkins University**

Mar 2026 - Present | Supervisor: Samuel Miles

- Contribute to a health-facility electricity reliability and medical-oxygen access study across three provinces of the Democratic Republic of the Congo.
- Support facility classification, outage and voltage analysis, oxygen-system vulnerability assessment, cross-province visualizations, predictive reliability modeling, and manuscript deliverables.
- Translate heterogeneous survey, facility, and device-energy data into reproducible research outputs for global-health infrastructure planning.

**Research Intern, Findability Sciences**

Aug 2025 - Dec 2025 | Advisors: Fatemeh Emdad, Chun-Kit Ngan, Bahman Moraffah; sponsors: Sagar and Prachi Mahurkar

- Built an industrial sensor-optimization project for sugarcane milling, reducing 150+ sensor variables into interpretable control features and operational recommendations.
- Produced Pareto-optimized recommendations associated with approximately 4.8% simulated efficiency improvement and presented the work as a research poster.
- Recognized as Most Valuable Player for the WPI Data Science Graduate Qualifying Project.

**Research Intern, thyssenkrupp Materials Services**

Jan 2025 - May 2026 | Collaborators: Chun-Kit Ngan and Rolf Bardeli

- Developed applied optimization research for rectangular and 3D packing under industrial constraints, connecting academic benchmark performance with practical logistics settings.
- Prepared experimental results, technical writeups, and extension materials for peer-reviewed dissemination.

**Graduate Student Ambassador, Worcester Polytechnic Institute**

Feb 2025 - Present

- Support prospective and admitted graduate students through program outreach, onboarding, campus tours, virtual sessions, and graduate community engagement.

## SELECTED PUBLICATIONS, MANUSCRIPTS, AND POSTERS

- A Multimodal Decision-Support Framework for Cardiology Triage with Learning-Based Test Selection and Explainable AI - Computing in Cardiology 2026, accepted regular track - Sam Guessoum et al.

- Automated Surgical Report Generation via Retrieval-Augmented LLMs for Cataract Surgery Video Understanding - SPIE Medical Imaging 2027, manuscript in preparation - Sam Guessoum, Mohammad Eslami et al.
- An Adaptive Multi-Agent System with Data-Driven Heuristic Orchestration for Solving 1D/2D Bin-Packing Problems - peer-reviewed conference paper; Springer LNCS-related extension - Sam Guessoum, Chun-Kit Ngan, Rolf Bardeli.
- UPMO: An Uncertainty-Aware Predictive Multi-Objective Optimization Framework for Smart Agriculture - IEEE ACIT 2026, accepted - Sam Guessoum, Naima Boukhiar.
- Towards Real-Time Precision Agriculture: A Task-Aware Multi-Modal Decision Framework - IEEE ISAIA 2026, accepted - Sam Guessoum, Naima Boukhiar.
- NeuroGraphMamba: A Spatial-Temporal Graph State-Space Architecture for Patient-Independent Epileptic Seizure Detection - AMAI 2026, accepted - Ikram Aissiou, Naima Boukhiar, Sam Guessoum, Riad Fellah.
- Head-and-neck tumor segmentation, TN staging, and recurrence-free survival prediction - HECKTOR 2026 Challenge, accepted - Sam Guessoum et al.
- Cardiology decision support and multimodal clinical AI research - AFRICAI/MICCAI Workshop 2026, accepted - Sam Guessoum et al.
- Data-Driven Optimization of Sugarcane Milling Processes - WPI Data Science GQP research poster, 2025 - Sam Guessoum et al.

## SELECTED PROJECTS

- Cataract Surgery Video Intelligence: interactive research system for surgical-video timeline construction, report generation, grounded Q&A;, uncertainty flags, and reviewer-facing export.
- Cardiology Triage Decision Support: multimodal research system evaluated on approximately 500 cardiologist-annotated ECGs and public cardiac benchmarks.
- Brain-Computer Interface Surgical Guidance: decision-support application for high-channel micro-ECoG streams and cortical-placement review.
- Health-Facility Electricity Reliability and Oxygen Access: global-health systems-analysis project with Johns Hopkins RISE Lab.

## TECHNICAL SKILLS

**Programming:** Python, JavaScript, SQL, HTML/CSS, R. **ML/AI:** PyTorch, scikit-learn, pandas, NumPy, computer vision, deep learning, reinforcement learning, statistical modeling, explainable AI. **Research:** experimental design, cross-validation, ablation studies, error analysis, Git, Streamlit, FastAPI, LaTeX, data visualization.

**Languages:** Arabic, English, and French fluent; Spanish intermediate.

## TRAINING, HONORS, LEADERSHIP, AND SERVICE

- Introduction to Computational Neuroscience Summer School, Arabs in Neuroscience, 2026.
- 5th Surgical Data Science Summer School, IHU Strasbourg / University of Strasbourg, 2026.
- Connecting Your IoT Devices, European Digital UniverCity / ENSSAT / University of Rennes, 2026.
- Winner, Tech for Mobility: Transforming Sidi Abdellah Transport Future, University of Algiers 1, 2025.
- Winner, AfriDev Hackathon, Worcester Polytechnic Institute, 2025.
- Winner, Hacking the Racial Wealth Gap: Jobs & Entrepreneurship, MIT Community Innovators Lab and partners, 2025.
- Best Paper recognition; MVP recognition; MICCAI Mentorship Program participant; AFRICAI/MICCAI workshop reviewer.
- Co-Founder, Octadot; WPI Graduate Student Ambassador; academic tutor supporting graduate-school and GRE preparation.